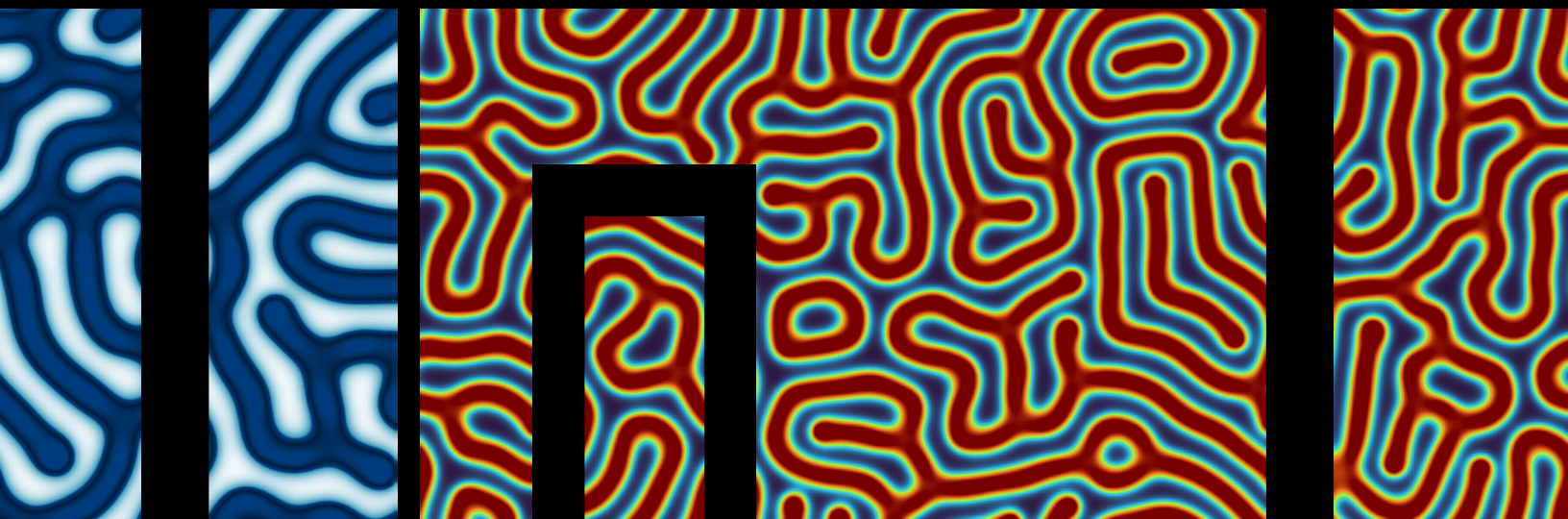




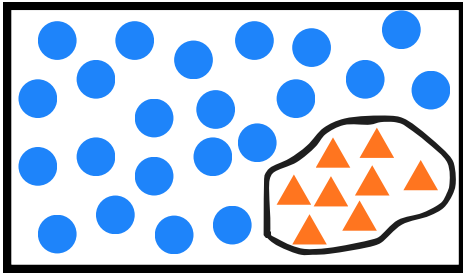
ALAN TURING

Morphogenesis

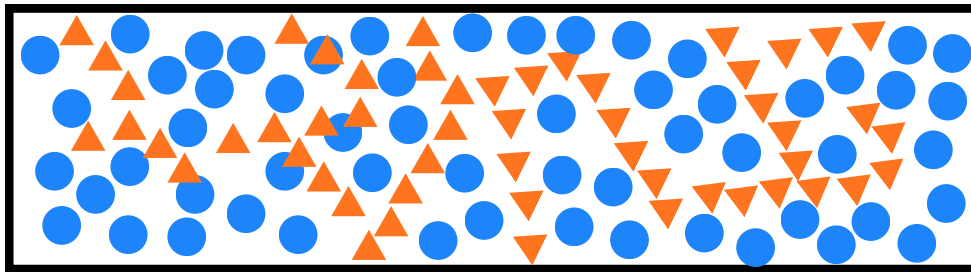
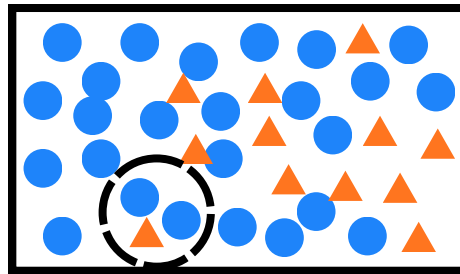
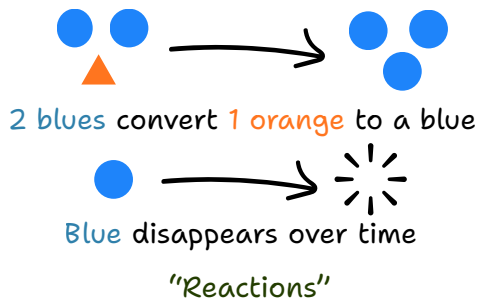
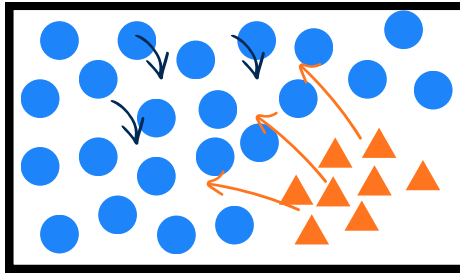
LIFE

turing pattern

Let's start with two chemicals



Orange diffuses faster than blue
i.e. Orange moves faster than blue



When diffusion and reactions take place together patterns form

Complex patterns can emerge from simple rules when chemicals or particles move and interact.

In 1952, **Alan Turing** asked a simple but powerful question: **How do living things know where to place spots, stripes, and shapes?**

He introduced the idea of **morphogenesis**: the process by which patterns and forms develop in living systems. Instead of assuming a detailed blueprint, Turing showed that order can emerge naturally from simple physical and chemical rules.

One such rule is called **reaction-diffusion**. In this process, chemicals react with one another as they spread through space at different speeds. A small imbalance can grow while nearby regions are suppressed, leading to stable patterns such as spots, stripes, or waves. These patterns are not pre-formed in advance they self-organize.

Beyond chemical reactions, additional patterning can arise through **diffusiophoresis**. In many systems, reaction-diffusion first creates chemical gradients (smooth changes in concentration from one place to another), the same gradients that give rise to Turing patterns.

Diffusiophoresis then acts on top of these gradients, causing particles to move in response to them. Even without particles reacting directly, their motion is guided by these reaction-generated concentration differences, leading to enhanced or secondary patterns.

Together, these ideas show how **simple local interactions, combined with motion and diffusion, can generate the rich patterns we see in nature.**

Scan the QR codes for deeper readings on pattern formation and our research group's work.



Our Group



Read more about this work in NYT

I am Dr. Turing, to learn more about me scan the QR below



Remembering Dr. Alan Turing - Wired

